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Hierarchical configuration decisions in transfer line balancing: Operational, tactical, and strategic effects on cycle time

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ABSTRACT

The Transfer Line Balancing Problem (TLBP) concerns the optimal assignment of tasks to workstations in automated machining lines to improve operational efficiency. This problem is particularly relevant to high-volume manufacturing industries, such as the automotive and aerospace industries, because it directly affects productivity and cost efficiency. This study examines how hierarchical configuration decisions at the operational (task assignment), tactical (joint task–tool assignment), and strategic (joint task–tool–fixture assignment) levels affect cycle time in an existing automotive machining transfer line. Three mixed-integer linear programming (MILP) models are developed to represent these configuration levels while accounting for setup times and industrial constraints, including task–machine accessibility, zoning, and technological restrictions. Because the exact models do not scale effectively to large instances, a constructive heuristic combined with a local search strategy is also developed. The proposed solution methods are evaluated using an industrial case study and 225 benchmark instances generated using a MILP-based benchmark generator. Across the benchmark size groups, operational cycle times are, on average, up to 11.87% and 12.65% higher than those obtained under tactical and strategic configurations, respectively, indicating potential productivity and economic benefits. The findings support managers in selecting an appropriate configuration level. Sensitivity analyses are also conducted to assess the effects of task–machine accessibility and task-time variability on solution performance.

1. Introduction

Transfer lines are highly automated, flow-oriented production systems widely used in high-volume manufacturing industries, including automotive and aerospace (Bratcu et al., 2005; Guschinskaya and Dolgui, 2009). Consisting of sequentially arranged workstations linked by automated material handling devices, these systems are designed for the mass production of identical or similar products with high precision, repeatability, and low per-unit cost (Guschinskaya and Dolgui, 2006; Dolgui et al., 2021). Each workstation is equipped with specialized machines, either multi-spindle heads or mono-spindle Computer Numerical Control (CNC) machines.

Transfer lines can consolidate multiple machining operations into fewer workstations, reducing equipment redundancy and cycle time and making them indispensable for competitive, large-scale manufacturing (Essafi et al., 2010b; Delorme et al., 2009). While transfer lines offer numerous benefits, their efficiency is significantly dependent on the ability to address several challenges at different configuration levels that require careful optimization. A key challenge is line balancing, which involves distributing tasks among different workstations while considering multiple constraints to optimize objectives such as cycle time, cost, or the number of machines (Battaïa and Dolgui, 2013; Battaïa and Dolgui, 2022; Abdous et al., 2025). Although line balancing has traditionally been associated with manual assembly systems, it is equally critical in transfer lines due to their unique machining characteristics (Beldar et al., 2025), giving rise to the Transfer Line Balancing Problem (TLBP) (Dolgui et al., 1999).

This study addresses the TLBP in serial transfer lines equipped with mono-spindle CNC machines, motivated by a real-world case from the automotive industry involving engine cylinder head production. While multi-spindle configurations have traditionally dominated TLBP research, modern high-precision manufacturing increasingly relies on

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mono-spindle CNC systems due to their flexibility, programmable tool magazines, and easier maintenance. Despite growing research on transfer line configurations, no prior work has systematically examined how hierarchical configuration decisions at the operational (task assignment), tactical (joint task–tool assignment), and strategic (joint task–tool–fixture assignment) levels influence cycle time in existing serial machining lines. This gap is particularly consequential because, in industrial practice, the scope of reconfiguration a manufacturer can undertake is constrained by time, cost, and organizational readiness, naturally structuring decisions into these three hierarchical levels. Importantly, this hierarchical approach enables cycle time reduction without requiring any modifications to the physical line layout, making it practically viable for manufacturers seeking efficiency gains within existing infrastructure. To the best of the authors' knowledge, this study is the first to systematically integrate and analyze such hierarchical configuration decisions in the TLBP literature.

To address this gap, this study makes the following contributions:

1. It introduces hierarchical configuration decisions into the TLBP by considering operational, tactical, and strategic configuration levels within serial single-model machining lines, and develops MILP models to minimize cycle time under each level.
2. It proposes a constructive heuristic to address the scalability limitations of the exact MILP approach, especially for tactical and strategic configurations where exact methods fail to solve most instances.
3. It generates benchmark datasets with different problem characteristics and analyzes the effects of hierarchical configuration decisions, task–machine accessibility, and task-time variability through a real case study and instances of different sizes.

The remainder of this paper is organized as follows: Section 2 reviews the literature on TLBP and existing methodologies. Section 3 defines the problem, outlines real-world constraints, and presents the mathematical formulation for different configuration levels. Section 4 details the proposed heuristic solution for the problem. Section 5 presents the computational experiments that validate the proposed methodology through a case study and the generated instances of various sizes. Managerial insights derived from the findings are discussed in Section 6, providing practical implications for decision-makers. Finally, conclusions and directions for future research are presented in Section 7.

2. Literature review

The Transfer Line Balancing Problem (TLBP) was initially introduced by Dolgui et al. (1999) for the mass production of a single product type, representing a significant shift in line balancing research from traditional assembly processes to machining operations. In the typical TLBP scenario, production lines function in a paced and sequential manner, with workstations linked by automated material handling systems (Dolgui et al., 2006b). Components are transferred smoothly between stations to maintain a continuous production flow (Osman and Baki, 2014). Each workstation is equipped with specialized tooling capable of performing machining tasks using either mono- or multi-spindle heads (Battaïa et al., 2023). Over the past two decades, TLBP has been a critical area of research, addressing the efficient allocation of tasks across workstations to optimize production efficiency while adhering to various real-world constraints (Petersen et al., 2024).

A recent study by Beldar et al. (2025) introduces a tuple notation framework ($\alpha \mid \beta \mid \gamma$) that systematically classifies TLBPs based on three critical dimensions: system configuration (α), problem characteristics (β), and optimization objectives (γ). In this study, we adopt this classification framework to position our research within the existing body of knowledge and to highlight our contributions to advancing the field.

2.1. Transfer line configurations: serial vs. parallel–serial

TLBP research can be broadly organized into two categories based on line topology: serial lines, where workstations are arranged in a single sequential chain, and parallel–serial lines, where multiple machines can operate in parallel at one or more stages. This distinction is critical as it reflects fundamentally different manufacturing contexts, optimization objectives, and technological choices, and shapes the methodological approaches used in the literature.

2.1.1. Serial single-model transfer lines

Table 1 provides a structured overview of key studies addressing serial single-model transfer lines (S1). A clear pattern emerges from the literature: research in this area has been almost exclusively focused on multi-spindle head configurations, with mono-spindle CNC systems remaining significantly under-studied despite their growing industrial relevance.

2.1.1.1. S1 with multi-spindle heads. The foundational work by Dolgui et al. (1999) established the TLBP for serial lines equipped with multi-spindle heads, minimizing line cost while satisfying precedence relationships and zoning constraints between tasks. Multi-spindle heads, which simultaneously execute multiple machining operations in parallel, were the dominant technology in high-volume automotive manufacturing at the time, and this naturally directed early research attention toward this configuration.

Over the following two decades, the multi-spindle serial line literature grew substantially, generating a rich variety of solution approaches. Mathematical programming formulations and graph-based methods were among the earliest contributions (Dolgui et al., 1999, 2000, 2004, 2006a; Guschinskaya and Dolgui, 2006), followed by heuristic approaches (Dolgui et al., 2003, 2005a; Guschinskaya and Dolgui, 2006; Dolgui et al., 2008), reduction techniques (Guschinskaya et al., 2007; Borisovsky et al., 2011; Battaïa and Dolgui, 2012), and set-partitioning formulations (Dolgui and Ihnatsenka, 2009; P. Borisovsky et al., 2012). Metaheuristic approaches were also developed, including greedy randomized adaptive search procedures (Guschinskaya and Dolgui, 2007; Guschinskaya et al., 2011), genetic algorithms (Guschinskaya et al., 2011), and combinatorial methods (Delorme et al., 2012). While most of these studies address the same core problem structure, some extended the model to incorporate practical considerations such as loading and unloading times (Guschinskaya et al., 2008, 2011), thereby enhancing the industrial realism of the formulations.

More recent studies have extended the multi-spindle serial line problem toward different objectives. Pirogov et al. (2021) examine service rate maximization, while Telemeci and Azizoğlu (2024) focus on cycle time minimization rather than line cost, reflecting the growing industrial emphasis on throughput performance. Despite this evolution in objectives, all multi-spindle serial line studies in the literature address exclusively operational-level decisions, namely the assignment of tasks to machines under a fixed equipment configuration. The configuration of tools and fixtures is treated as given and immutable in all these research works.

2.1.1.2. S1 with mono-spindle heads. In contrast to the rich multi-spindle literature, serial transfer lines with mono-spindle CNC machines have received very limited research attention. Mono-spindle machines execute operations sequentially rather than in parallel, and their programmable tool magazines, multi-face machining capabilities, and relative ease of maintenance make them increasingly prevalent in modern high-precision manufacturing contexts such as engine component production.

Nourmohammadi et al. (2024) studied a mono-spindle serial line that incorporates a comprehensive set of constraints, including technological compatibility, task–machine accessibility, tool change times, table rotation times, and scheduling decisions, with the dual objective of minimizing cycle time and energy consumption. This work represents

Table 1
Key studies focusing on S1.

Authors	Notation	Approach
Dolgui et al., 1999	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	MM , G
Dolgui et al., 2000	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	MM , G
Dolgui et al., 2003	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	MM , H
Dolgui et al., 2004	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	MM , H
Dolgui et al., 2005a	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	H
Dolgui et al., 2005b	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	H
Guschinskaya and Dolgui, 2006	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	MM , H
Guschinskaya et al., 2007	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	RM
Guschinskaya and Dolgui, 2007	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	GRASP
Dolgui et al., 2008	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	G, MM, H
Guschinskaya et al., 2008	$S1 \mid C^{prec}, C^{inc}, C^{exc}, LU^{ti}, SH^{multi} \mid Li^{co}$	H
Dolgui and Ihnatsenka, 2009	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	SP
Borisovsky et al., 2011	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	RM
Guschinskaya et al., 2011	$S1 \mid C^{prec}, C^{inc}, C^{exc}, LU^{ti}, SH^{multi} \mid Li^{co}$	GRASP , GA
P. Borisovsky et al., 2012	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	SP
Delorme et al., 2012	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	CA
Battaïa and Dolgui, 2012	$S1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{multi} \mid Li^{co}$	RM
Pirogov et al., 2021	$S1 \mid C^{prec}, SH^{multi} \mid SR$	MM , H
Telemeci and Azizoğlu, 2024	$S1 \mid C^{prec}, C^{exc}, SH^{multi} \mid Cyc^{ti}$	MM , B&B
Nourmohammadi et al., 2024	$S1 \mid C^{prec}, C^{exc}, C^{tec}, C^{inc}, C^{acc}, SH^{mono}, TC^{ti}, TaR^{ti}, Sch \mid Cyc^{ti}, En^{co}$	MM
Beldar et al., 2024	$S1 \mid C^{prec}, C^{exc}, C^{inc}, C^{acc}, SH^{mono}, TC^{ti}, TaR^{ti} \mid Cyc^{ti}$	H

Abbreviations – Solution approaches: Mathematical model (MM), Graph approach (G), Branch and Bound (B&B), Heuristic (H), Reduction method (RM), Greedy randomized adaptive search procedure (GRASP), Set partitioning (SP), Genetic Algorithm (GA), Combinatorial approach (CA). **Problem characteristics:** Precedence constraints (C^{prec}), Inclusion constraints (C^{inc}), Exclusion constraints (C^{exc}), Accessibility constraints (C^{acc}), Technological constraints (C^{tec}), Tool change time (TC^{ti}), Table rotation time (TaR^{ti}), Loading/unloading time (LU^{ti}), Scheduling (Sch), Multi-spindle head (SH^{multi}), Mono-spindle head (SH^{mono}). **Objectives:** Line cost (Li^{co}), Cycle time (Cyc^{ti}), Service rate (SR), Energy consumption (En^{co}).

an important advance in capturing the time-related complexity that is distinctive to sequential mono-spindle operations. Beldar et al. (2024) subsequently proposed a heuristic for a closely related mono-spindle configuration, focusing on cycle time minimization under precedence, zoning, accessibility, and time-related constraints.

Both studies make valuable contributions to the mono-spindle serial line literature. However, both treat the tool and fixture configuration of each machine as fixed and address only operational-level task assignment decisions. Neither considers the possibility of reallocating tools or fixtures across machines as part of the optimization, leaving a significant dimension of the problem unexplored.

2.1.2. Parallel–serial single-model transfer lines

Table 2 provides an overview of key studies on parallel–serial single-model transfer lines (PS1), where each stage may contain multiple machines operating in parallel. A notable characteristic of this literature is that the reviewed parallel–serial studies employ mono-spindle CNC machines, reflecting the modern trend toward flexible, programmable architectures that accommodate product variety and routing alternatives.

Early research in this area (Essafi et al., 2009, 2010c,a, 2012) addresses line design problems that simultaneously determine the number and configuration of workstations and assign tasks to machines, with the objective of minimizing line cost or the number of machines. These studies incorporate accessibility constraints and sequence-dependent setup times, and are solved using greedy randomized adaptive search, mathematical programming, and ant colony optimization. Subsequent contributions by P.A. Borisovsky et al. (2012), Borisovsky et al. (2013, 2014) developed more advanced solution techniques, including dynamic programming, genetic algorithms, column generation, and branch-and-cut, for parallel–serial design problems with sequencing constraints.

Research in this stream progressively incorporated more complex operational elements. Osman and Baki (2014, 2018) examined configurations that account for tool assignment, tool magazine capacity, task sequencing, and a range of time-related factors, including tool change times, orientation change times, and loading and unloading operations, optimizing non-productive time and manpower costs. Liu et al. (2016) introduced a multi-objective model balancing the number of machines, cycle time, and line balance rate under tool magazine capacity constraints and sequence-dependent setup times.

A particularly notable contribution is that of Shao et al. (2020), who explicitly incorporate joint tool and fixture assignment decisions, along with buffer allocation, into a multi-objective optimization framework that minimizes tool cost and maximizes product rate, solved using metaheuristic and simulation-based methods. This work represents the most comprehensive treatment of tool and fixture decisions in the parallel–serial transfer line literature. However, as with all parallel–serial studies, these decisions are embedded within a new line design problem, where the parallel routing architecture itself is subject to optimization, a fundamentally different setting from reconfiguring an existing serial line.

Recent parallel–serial contributions have continued to broaden the scope of the problem. Energy consumption has also been incorporated as an objective (Delorme et al., 2023; Delorme and Gianessi, 2024). Manufacturing and reconfiguration costs are addressed by Liu et al. (2021). Scalability is examined by Cerqueus and Delorme (2023), and service level considerations appear in Delorme et al. (2023). Across all of these works, the focus remains on designing or reconfiguring the architecture of parallel–serial lines rather than improving the operational and configurational performance of a fixed serial line.

2.2. Research gaps

Table 3 summarizes the key conceptual differences between parallel–serial line design studies and the present work, providing a structured basis for positioning our contributions within the existing literature.

As shown in Table 3, the present work differs fundamentally from parallel–serial line design studies in both scope and intent. Prior parallel–serial research optimizes the architecture of new lines by

Table 2
Key studies focusing on PS1.

Authors	Notation	Approach
Essafi et al., 2009	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid Li^{co}$	GRASP
Essafi et al., 2010c	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid NM$	MM
Essafi et al., 2010a	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid Li^{co}$	ACO
P.A. Borisovsky et al., 2012	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, Seq, SH^{mono} \mid NM$	DP
Essafi et al., 2012	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid Li^{co}$	GRASP
Borisovsky et al., 2013	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid NM$	GA
Borisovsky et al., 2014	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid NM$	CG, B&C, parallel B&C
Osman and Baki, 2014	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, A^{to}, To^{ma}, Seq, TC^{ti}, Re^{ti}, Tr^{ti}, OC^{ti}, SH^{mono} \mid NonP^{ti}$	MM, D, D+ACO, ACO
Delorme et al., 2016	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid Li^{co}, PR$	MM, GRASP
Liu et al., 2016	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, To^{ma}, TC^{ti}, OC^{ti}, Sd^{st}, SH^{mono} \mid NM, Cyc^{ti}, LBR$	MM, H
Osman and Baki, 2018	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, A^{to}, Ac^{C}$ $To^{ma}, Seq, TC^{ti}, LU^{ti}, OC^{ti}, SH^{mono}$ $\mid NonP^{ti}, Man^{co}$	MM, CS
Lahrichi et al., 2019	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid NM$	PA
Borisovsky, 2020	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, SH^{mono} \mid NM$	GA
Shao et al., 2020	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, A^{to}, A^{f}, Bu^{al}, SH^{mono} \mid To^{co}, PR$	MM, MOPSO, NSGA-II, Sim
Liu et al., 2021	$PS1 \mid C^{prec}, Ac^{C}, SH^{mono} \mid Man^{co}, Rec^{co}$	MM, GRASP
Lahrichi et al., 2021a	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, Seq, SH^{mono} \mid NM$	MH
Lahrichi et al., 2021b	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, Ac^{C}, Sd^{st}, Seq, SH^{mono} \mid NM$	PA
Cerqueus and Delorme, 2023	$PS1 \mid C^{prec} \mid Scal$	CA
Delorme et al., 2023	$PS1 \mid C^{prec} \mid Li^{co}, En^{co}, Se^{le}$	MH
Borisovsky, 2023	$PS1 \mid C^{prec}, C^{inc}, C^{exc}, SH^{mono}, Sd^{st} \mid NM$	GA
Delorme and Gianessi, 2024	$PS1 \mid C^{prec}, Sch \mid En^{co}$	MM, MH

Abbreviations – Solution approaches (new): Decomposition approach (D), Branch and Cut (B&C), Dynamic programming (DP), Cuckoo search (CS), Constraint generation (CG), Ant colony optimization (ACO), Polynomial algorithm (PA), Simulation (Sim), Matheuristic (MH). **Problem characteristics (new):** Accessibility constraints (Ac^{C}), Sequence-dependent setup times (Sd^{st}), Tool assignment (A^{to}), Tool magazine capacity (To^{ma}), Sequencing (Seq), Refixturing or reloading time (Re^{ti}), Transfer time (Tr^{ti}), Orientation change time (OC^{ti}), Fixture assignment (A^{f}), Buffer allocation (Bu^{al}). **Objectives (new):** Number of machines (NM), Non-productive time ($NonP^{ti}$), Manpower cost (Man^{co}), Tool cost (To^{co}), Product rate (PR), Reconfiguration cost (Rec^{co}), Line balance rate (LBR), Service level (Se^{le}), Scalability ($Scal$).

Table 3
Conceptual positioning: Prior line design studies vs. this work.

Dimension	Parallel-serial line design studies	This work
Primary focus	New line architecture design	Reconfiguration of an existing serial line
Decision scope	Line topology + task assignment	Task, tool, and fixture assignment only
Line topology	Determined by optimization	Fixed (existing infrastructure)
Configuration levels	Single-level design decision	Three hierarchical levels: operational, tactical, and strategic
Planning horizon	Long-term capital investment	Days to months, depending on level
Investment scale	High (new installation)	Low to moderate (reconfiguration within existing line)

simultaneously determining the number, arrangement, and configuration of workstations. This paper, by contrast, targets the performance improvement of an already-installed serial transfer line with a fixed sequential topology, where reconfiguration must occur entirely within the existing physical infrastructure.

Three specific gaps in the literature motivate this study:

(1) Multi-level reconfiguration within a fixed topology. No existing study, whether on serial or parallel–serial lines, examines how configuration decisions can be structured across multiple levels of scope within a fixed serial line. The present work introduces three hierarchical levels: operational (task assignment only), tactical (joint task and tool assignment), and strategic (joint task, tool, and fixture assignment). Each level expands the configuration space and unlocks further cycle time reductions without requiring any modification to the physical line layout.

(2) Trade-off between implementation effort and performance gain. Prior studies do not differentiate reconfiguration decisions according to their practical implementation complexity, nor do they analyze how increasing reconfiguration effort influences achievable cycle time improvement. In practice, task reassignment can often be implemented rapidly with limited disruption, whereas tool reallocation and fixture reconfiguration require progressively greater time, coordination, and investment. This study explicitly captures that progression and links each configuration level to a different implementation burden.

(3) Integration with real-world mono-spindle constraints. No prior work integrates such hierarchical configuration decisions with the comprehensive set of practical constraints that characterize real-world mono-spindle operations, including task–machine accessibility and technological compatibility restrictions, detailed time components such as tool changes, table rotations, spindle movements, and loading/unloading operations, and explicit fixture–task and machine–fixture relationships.

This paper addresses all three gaps by formulating MILP models and a constructive heuristic for each configuration level, and evaluating their impact on cycle time through a real-world automotive case study and a set of generated benchmark instances. The full problem formulation and mathematical models are presented in Section 3.

3. Problem definition

The TLBP investigated in this study is defined for a transfer line comprising a series of stations connected by an overhead crane, where each station is equipped with advanced CNC machines featuring mono-spindle heads and tool magazines, fixtures, and rotary tables. At each machine, fixtures clamp the part, and a rotary table orients it to the required machining side. The tool magazine enables the machine to change to the required tool. This study examines TLBP in an existing machining line, where decision-makers aim to balance machining

Table 4
Summary of problem considerations.

Category	Description
Equipment	Tool assignment: required tools are allocated per operation to minimize tool change delays. Fixture assignment: fixtures are allocated per machine considering task requirements and task–fixture relationships. Tool magazine capacity: total tool slots must not exceed magazine capacity. Mono-spindle head: operations are performed sequentially, requiring careful task ordering.
Time	Loading/unloading time: applied at the first and last task of each machine. Movement to task position: spindle travel time to task location. Movement back to tool change position: spindle return time after task completion. Machining time: processing time for each task. Table rotation time: time to rotate the part between machining sides. Extra measurement time: calibration overhead when linked tasks are performed on different machines. Tool change time: time to switch tools between consecutive tasks.
Assignment	Precedence: tasks must follow a technologically correct sequence. Zoning: inclusion requires co-location; exclusion prevents co-location. Accessibility: feasible task–machine assignments depend on the configuration level. Technological compatibility: certain tasks are restricted to specific machines. Sequencing: task execution order within each machine, considering tool changes, rotations, and loading/unloading.

operations to minimize cycle time while considering hierarchical configuration decisions. At the operational level, task assignment decisions are made by adjusting only task distribution among machines according to the current configurations of tools and fixtures. At the tactical level, both task and tool assignment decisions are optimized by redistributing tasks and tools across machines, assuming fixtures remain unchanged. At the strategic level, comprehensive configuration decisions are achieved by simultaneously optimizing task, tool, and fixture distributions across machines. As decision-makers transition from operational to strategic levels, the planning horizon ranges from days to months or longer, and implementation complexity increases from minor adjustments to substantial reconfigurations of the existing line. In addition to these hierarchical configuration decisions, the problem incorporates several critical real-world considerations, as categorized and discussed below.

3.1. Problem considerations

Table 4 summarizes the equipment-related, time-related, and assignment-related considerations incorporated in this study.

3.2. Mathematical model

3.2.1. Modeling overview

The MILP model presented in this section minimizes cycle time by assigning tasks to machines while satisfying all constraints described above. The model is structured in three incremental layers corresponding to the three configuration levels, as summarized in Table 5. The operational model forms the base layer, capturing the fundamental complexity of the problem under fixed tool and fixture configurations. The tactical model extends this by introducing a tool assignment decision variable and relaxing machine accessibility accordingly. The strategic model further extends the tactical model by adding a fixture assignment decision variable, achieving the broadest configuration scope.

The cycle time computation is the central modeling challenge. For each machine, the completion time accumulates three components: (1) the sum of task processing and spindle movement times, (2) transition times between consecutive tasks accounting for tool changes and table rotations, and (3) loading and unloading times determined by the first and last tasks assigned to each machine. Identifying the first and last tasks on each machine requires a linearization procedure involving auxiliary binary and continuous variables. To preserve the readability of the main model, this linearization is presented in full in Appendix A.

3.2.2. Operational level model

Table 6 presents the sets, parameters, and decision variables at the operational level.

$$\text{Minimize} \quad CT \tag{1}$$

$$\sum_{j\in J,o\in O} x_{ijo} = 1, \qquad \forall i \in I \tag{2}$$

$$\sum_{i\in I} x_{ijo} \le 1, \qquad \forall j \in J, o \in O \tag{3}$$

$$\sum_{i\in I,o\in O} x_{ijo} \ge 1, \qquad \forall j \in J \tag{4}$$

$$\sum_{j,o} \left(j + \frac{o}{NOR}\right) x_{ijo} \le \sum_{j,o} \left(j + \frac{o}{NOR}\right) x_{i'jo}, \qquad \forall i, i' \in I \text{ if } Pre_{i,i'} = 1 \tag{5}$$

$$\sum_{i\in I} x_{ijo} \ge \sum_{i\in I} x_{ij,o+1}, \qquad \forall j \in J, o < NOR \tag{6}$$

$$mc_j = \sum_{i,o} x_{ijo} \cdot \left(MTP_i + t_i + MBTP_i\right) + \sum_{i\neq i',o} w_{ii'jo} \cdot \max\left(TR_{s,s'}, TC_t_{i,i'} \cdot TC_j\right), \quad \forall j \in J,\ s \in TS_i,\ s' \in TS_{i'} \tag{7}$$

$$w_{ii'jo} \le x_{ijo}, \qquad \forall i, i' \in I, j \in J, o \in O,\ i \neq i',\ o < NOR \tag{8}$$

$$w_{ii'jo} \le x_{i'j,o+1}, \qquad \forall i, i' \in I, j \in J, o \in O,\ i \neq i',\ o < NOR \tag{9}$$

Table 5
Structure of MILP models across configuration levels.

Element	Operational	Tactical	Strategic
Task assignment	Decision variable	Decision variable	Decision variable
Tool assignment	Input parameter	Decision variable	Decision variable
Fixture assignment	Input parameter	Input parameter	Decision variable
Accessibility matrix	$Acc1$	$Acc2$	$Acc3$
Additional decision variable	–	toa_{ij}	fia_{fj}
Constraints	(1)–(23)	+ (24)–(25)	+ (26)–(27)
Planning horizon	Days	Weeks to months	Months to a year
Implementation complexity	Low	Moderate	High

Table 6
Sets, parameters, and decision variables of the operational level model.

Sets	**Description**
I	Set of tasks, indexed by i, where $\|I\| = NT$
J	Set of machines, indexed by j, where $\|J\| = NM$
O	Set of task-order positions, indexed by o, where $\|O\| = NOR$
T	Set of tools, indexed by t, where $\|T\| = NTO$
F	Set of fixtures, indexed by f, where $\|F\| = NF$
S	Set of machining sides, indexed by s, where $\|S\| = NS$
Parameters	**Description**
B	Big-M constant (see Appendix A)
δ	Small constant for linearization
MTP_i	Movement to position time of task i
t_i	Processing time of task i
$MBTP_i$	Movement back to tool change position time of task i
TC_j	Tool change time on machine j
LD_j	Loading time of machine j
LS_j	Loading side of machine j
m_j	Maximum number of tasks allowed on machine j
$TR_{s,s'}$	Table rotation time from side s to side s'
$Ext_{i,i'}$	Extra measurement time between tasks i and i'
$Pre_{i,i'}$	Precedence: 1 if task i must precede task i', 0 otherwise
$Zp_{i,i'}$	Inclusion: 1 if tasks i and i' must be on the same machine, 0 otherwise
$Zn_{i,i'}$	Exclusion: 1 if tasks i and i' cannot be on the same machine, 0 otherwise
$Ter_{i,j}$	Technological: 1 if task i must be performed on machine j, 0 otherwise
$Pre_t_f_{i,f}$	Task–fixture incompatibility: 1 if task i cannot be performed with fixture f, 0 otherwise
$Mf_{j,f}$	Machine-fixture: 1 if machine j uses fixture f, 0 otherwise
TS_i	Side index of task i
TTo_i	Tool index of task i
$TC_t_{i,i'}$	1 if a tool change is needed between tasks i and i', 0 otherwise
$Acc1_{ij}$	Accessibility of task i on machine j at the operational level: 1 if task i can be accessed on machine j, 0 otherwise
Decision variables	**Description**
$x_{ijo} \in \{0,1\}$	1 if task i is assigned to machine j in order o, 0 otherwise
$fl_{ii'j} \in \{0,1\}$	1 if task i is the first and task i' is the last task on machine j, 0 otherwise
$w_{ii'jo} \in \{0,1\}$	Auxiliary binary variable
$CT \geq 0$	Cycle time
$mc_j \geq 0$	Basic completion time of machine j
$mc2_j \geq 0$	Completion time of machine j (with extra measurement)
$mc3_j \geq 0$	Completion time of machine j (with loading and unloading)
$ntm_j \in \mathbb{Z}^+$	Number of tasks on machine j
$tor_{ij} \in \mathbb{Z}^+$	Order of task i on machine j

$$w_{ii'jo} \geq x_{ijo} + x_{i'j,o+1} - 1, \qquad \forall i,i' \in I, j \in J, o \in O,\ i \neq i',\ o < NOR \tag{10}$$

$$Ext_{i,i'} > 0,\ j < j' \implies mc_{j'} + Ext_{i,i'} \cdot \left(\sum_o x_{i'j'o} - \sum_o x_{ij'o} \right) \leq mc2_{j'}, \quad \forall i,i' \in I,\ j,j' \in J \tag{11}$$

$$Ext_{i,i'} > 0,\ j < j' \implies mc_{j'} - Ext_{i,i'} \cdot \left(\sum_o x_{i'j'o} - \sum_o x_{ij'o} \right) \leq mc2_{j'}, \quad \forall i,i' \in I,\ j,j' \in J \tag{12}$$

$$mc2_j \geq mc_j, \qquad \forall j \in J \tag{13}$$

$$\sum_{o \in O} x_{ijo} \leq Acc1_{ij}, \qquad \forall i \in I, j \in J \tag{14}$$

$$Pre_t_f_{i,f} = 1 \wedge Mf_{j,f} = 1 \implies \sum_o x_{ijo} = 0, \qquad \forall i,j,f \tag{15}$$

$$Zp_{i,i'} = 1 \implies \sum_o x_{ijo} = \sum_o x_{i'jo}, \qquad \forall i,i' \in I, j \in J \tag{16}$$

$$Zn_{i,i'} = 1 \implies \sum_o x_{ijo} + \sum_o x_{i'jo} \leq 1, \qquad \forall i,i' \in I, j \in J \tag{17}$$

$$Ter_{i,j} = 1 \implies \sum_o x_{ijo} = 1, \qquad \forall i \in I, j \in J \tag{18}$$

$$ntm_j = \sum_{i,o} x_{ijo}, \qquad \forall j \in J \tag{19}$$

$$tor_{ij} = \sum_o o \cdot x_{ijo}, \qquad \forall i \in I, j \in J \tag{20}$$

The variable $fl_{ii'j}$ identifies the first and last tasks on each machine and is used in Constraint (21) to compute loading and unloading contributions to the cycle time. Its derivation relies on a linearization procedure involving auxiliary variables $z1_{ii'j}$, $z2_{ii'j}$, and $stdi_{ii'j}$,

Table 7
Additional parameters and decision variable at the tactical configuration level.

Additional parameters	Description
To_slot_t	Number of magazine slots required by tool t
$TOMAG$	Tool magazine capacity
$Acc2_{ij}$	Accessibility of task i on machine j at the tactical level, determined by allowing the reassignment of tools
Additional decision variable	**Description**
$toa_{tj} \in \{0,1\}$	1 if tool t is assigned to machine j, 0 otherwise

Table 8
Additional parameters and decision variable at the strategic configuration level.

Additional parameters	Description
$Tafi_{if}$	Task–fixture requirement: 1 if task i requires fixture f, 0 otherwise
$Acc3_{ij}$	Accessibility of task i on machine j at the strategic level, determined by allowing the reassignment of tools and fixtures
Additional decision variable	**Description**
$fia_{fj} \in \{0,1\}$	1 if fixture f is assigned to machine j, 0 otherwise

presented in full in Appendix A.

$$mc2_j + \max\left(MBTP_{i'} + TC_t_{i,i'} \cdot TC_j + MTP_i,\ TR_{s,s'} + LD_j\right) \cdot fl_{ii'j} \le mc3_j, \quad \forall i,i' \in I,\ i \ne i',\ j \in J,\ s \in LS_j,\ s' \in TS_{i'} \tag{21}$$

$$mc3_j \le CT, \qquad \forall j \in J \tag{22}$$

$$\sum_{i,o} x_{ijo} \le m_j, \qquad \forall j \in J \tag{23}$$

The objective function (1) minimizes the cycle time. Constraints (2)–(4) assign each task to exactly one machine and order, allow at most one task per order, and require each machine to process at least one task. Constraint (5) enforces precedence, while Constraint (6) ensures sequential order assignment on each machine. Constraint (7) computes machine completion time mc_j from task processing, spindle movement, and the larger of table rotation or tool change times between consecutive tasks. Constraints (8)–(10) define the auxiliary variable $w_{ii'jo}$ to identify consecutive task pairs on the same machine. Constraints (11)–(13) account for additional measurement time between tasks on different machines. Constraints (14)–(18) enforce accessibility, task–fixture incompatibility, zoning, and technological requirements. Constraints (19)–(20) compute the number and positions of tasks on each machine. Constraint (21) adds loading and unloading time based on the first and last task on each machine, Constraint (22) defines the cycle time as the maximum machine completion time, and Constraint (23) enforces the maximum task capacity per machine.

3.2.3. Tactical level extension

At the tactical configuration level, tool assignments are jointly optimized with task assignments, allowing tools to be relocated between machines along with their associated operations. This relaxes the accessibility matrix from $Acc1$ to $Acc2$, substantially expanding the feasible assignment space (from 13% to 58% accessibility). Table 7 summarizes the additional parameters and decision variable at the tactical configuration level.

$$\sum_{o \in O} x_{ijo} \le toa_{tj}, \qquad \forall i \in I, j \in J, t \in T \text{ if } t = TTo_i \tag{24}$$

$$\sum_{t \in T} 2 \cdot toa_{tj} \cdot To_slot_t \le TOMAG, \qquad \forall j \in J \tag{25}$$

Constraint (24) ensures that if a task requiring a particular tool is assigned to a machine, that tool must also be assigned to that machine. Constraint (25) ensures that the total number of tool slots assigned to each machine does not exceed its magazine capacity; the factor of two accounts for tool duplication in the magazine. At this level, $Acc1_{ij}$ in Constraint (14) is replaced by $Acc2_{ij}$.

3.2.4. Strategic level extension

At the strategic configuration level, fixture assignments are additionally optimized alongside task and tool assignments, allowing fixtures to be relocated between machines along with their associated operations. This further relaxes the accessibility matrix from $Acc2$ to $Acc3$, expanding accessibility to 100%, meaning that all task–machine combinations are initially accessible, subject to the task–fixture requirements represented by $Tafi_{if}$. Table 8 summarizes the additional parameters and decision variable at the strategic configuration level.

$$\sum_{o \in O} x_{ijo} \le fia_{fj}, \qquad \forall i \in I, j \in J, f \in F \text{ if } Tafi_{if} = 1 \tag{26}$$

$$\sum_{f \in F} fia_{fj} = 1, \qquad \forall j \in J \tag{27}$$

Constraint (26) ensures that a task requiring a specific fixture can only be assigned to a machine if that fixture is assigned to the machine. Constraint (27) ensures that each machine is assigned exactly one fixture. At the strategic level, $Acc1_{ij}$ in Constraint (14) is replaced by $Acc3_{ij}$.

In the studied setting, the task–machine accessibility density increases from 13% at the operational level to 58% at the tactical level and 100% at the strategic level.

4. Solution method

While the mathematical formulation can find optimal or feasible solutions for small-scale instances at the operational level, it faces computational challenges for larger problems. Moreover, at the tactical and strategic configuration levels, the mathematical model can solve only small-scale test instances and fails to provide solutions for medium- and large-scale problems. To address these limitations and obtain feasible solutions within reasonable execution times across different configuration levels, a constructive heuristic capable of handling the various constraints is developed.

The proposed solution method follows a three-phase structure. In the first phase, a constructive heuristic generates a feasible solution for the operational configuration level by ordering tasks according to precedence relations and assigning them to machines while satisfying accessibility, zoning, technological, and fixture-related constraints. In the second phase, this operational solution is improved for the tactical level through a local search procedure that allows tool reallocation across machines. In the third phase, the resulting solution is further refined for the strategic level through a local search that allows both tool and fixture reallocation. This hierarchical structure reflects the three configuration levels of the TLBP and allows the method to remain computationally efficient while progressively increasing configuration flexibility.

The proposed heuristic methodology is outlined in the following steps:

1. *Task prioritization:* This step provides the initial task ordering by analyzing the precedence matrix ($Pre_{i,i'}$) to identify dependencies between tasks. The algorithm calculates a value ("in-degree") for each task based on the number of tasks that must precede it according to the precedence matrix. Tasks are then placed into a queue based on this value, prioritizing tasks with no dependencies (or those whose dependencies have already been processed). The algorithm proceeds by iteratively removing tasks from the queue and updating the dependencies of subsequent tasks until a refined task order is generated.
2. *Initial task assignment:* This step distributes tasks to all machines according to the order determined in the previous step. Initially, all machines receive the same set of tasks, establishing a standardized starting point for further refinement.
3. *Task–machine compatibility matching:* This step enforces task–machine compatibility by using the task–machine matrix (Acc_{ij}) to identify infeasible task assignments. For each machine and task combination, the matrix indicates whether the task can be performed on that specific machine. If a task is incompatible with a machine (indicated by a 0 in the matrix), the corresponding entry in the `TaskAssignment` matrix is set to −1, marking the task as infeasible for that machine. This filtering ensures that subsequent steps only consider feasible assignments.
4. *Positive zoning constraint implementation:* This step enforces the positive zoning constraints, ensuring that tasks that must be co-located are assigned to the same machine. The algorithm iterates through the positive zoning matrix ($Zp_{ii'}$). If two tasks are required to be on the same machine (indicated by a value greater than 0 in the matrix), the algorithm checks if both tasks are currently assigned to the same machine in the `TaskAssignment` matrix. If one task is assigned and the other is not, the unassigned task is marked as infeasible (−1) for that machine. This ensures that the co-location requirement is satisfied.
5. *Negative zoning constraint implementation:* This step enforces the negative zoning constraints, ensuring that tasks that must not be co-located are not assigned to the same machine. The algorithm iterates through the negative zoning matrix ($Zn_{ii'}$). If two tasks are prohibited from being on the same machine (indicated by a value greater than 0 in the matrix), the algorithm checks if both tasks are currently assigned to the same machine in the `TaskAssignment` matrix. If both tasks are found on the same machine, one of the tasks is removed from that machine, prioritizing tasks with higher dependencies. Additionally, to maintain exclusivity, the removed task is also marked as incompatible (−1) on any subsequent machine that initially has both tasks assigned.
6. *Technological constraint implementation:* This step enforces the technological constraints, ensuring that tasks are only assigned to machines that are technologically capable of performing them. The algorithm iterates through the technological matrix (Ter_{ij}). If a task must be performed on a specific machine (indicated by a 1 in the matrix), the algorithm ensures that the task is assigned only to that machine. If the task is assigned to any other machine, it is removed from that machine's assignment. This ensures that tasks adhere to technological compatibility requirements.
7. *Task–fixture incompatibility implementation:* This step ensures that tasks are not assigned to machines equipped with incompatible fixtures. The algorithm uses the task–fixture incompatibility matrix ($Pre_t_f_{i,f}$) to identify incompatible task–fixture combinations and the machine–fixture matrix (Mf_{jf}) to determine which fixture is installed on each machine. If $Pre_t_f_{i,f} = 1$ and $Mf_{jf} = 1$, task i is marked as infeasible (−1) for machine j. This ensures that only task–machine assignments with compatible fixture configurations are considered.
8. *Final assignment:* This step performs the final task assignment to machines, aiming to balance the workload while satisfying all constraints. First, a threshold is calculated by dividing the total task time by the number of machines and multiplying by a coefficient β ($\beta \geq 1$). The algorithm begins with the first machine and assigns tasks based on the prioritized task list. Before assigning a task, the algorithm checks if all prerequisites for that task have been completed. A task can only be assigned if all its prerequisites are already assigned to the same or earlier machines. The algorithm continues assigning tasks to the machine until the sum of the task times assigned to that machine does not exceed the calculated threshold. If the threshold is exceeded, the tasks assigned to the machine are marked as infeasible (−1) on all remaining machines, indicating that they are no longer eligible for assignment. Any tasks that cannot be assigned to the current machine without exceeding the cycle time are also marked as infeasible for that machine. In cases where a task must be performed on a specific machine, and there is no alternative machine available, the task is assigned to the designated machine, even if the cycle time threshold is exceeded. This process is repeated for each machine except the last one.
9. *Cycle time calculation:* This step calculates the cycle time for the entire transfer line. After the final assignment of tasks has been determined, the algorithm calculates the cycle time for each machine. The completion time of a machine is the sum of the processing times of all tasks assigned to it, plus any setup times (e.g., tool changes, table rotations, loading and unloading). The cycle time for the transfer line is then determined as the maximum cycle time across all machines. This value represents the time it takes for the entire production process to complete one unit of output and is the objective function that the heuristic seeks to minimize. A numerical example of the proposed heuristic is available at: https://figshare.com/s/4c754e4b7b8f31acdf6a

At the tactical configuration level, tools gain greater freedom of movement between machines compared to the more constrained operational level, based on the updated task–machine matrix ($Acc_{i,j}$). Using the feasible solution obtained at the operational level, we implement a local search strategy to further optimize the solution by specifically considering tool-based constraints. This approach aims to achieve shorter cycle times by enhancing tool utilization and improving machine accessibility. The local search systematically explores four predefined neighborhoods: (1) swapping tasks on the same machine, (2) transferring a task from the machine with the highest cycle time to any non-adjacent machine, (3) shifting a task from one machine to its adjacent neighbor, and (4) moving a task from the most loaded machine to a less loaded one. As detailed in Algorithm 1, the procedure iteratively evaluates one neighborhood at a time, checking all possible moves within it. If a move results in a feasible solution and reduces the cycle time, the best solution is updated, and the search restarts from the first neighborhood. Otherwise, the algorithm proceeds to the next neighborhood. This process continues until all neighborhoods are explored without further improvement, ensuring convergence to a local optimum.

At the strategic configuration level, the configuration decisions encompass tasks, tools, and fixtures, allowing all three elements to be reassigned across machines. This represents the most comprehensive configuration approach, where all tasks can potentially be assigned to all machines, subject to the task–fixture requirements and the associated tool and fixture allocation constraints. Building on the feasible solutions from the operational and tactical configuration levels, we reapply the local search strategy to further optimize assignments while addressing tool and fixture allocation constraints.

Algorithm 1: Local search strategy

Input: Initial feasible assignment, max_iterations
Output: Best assignment found (*best_assignment*), best cycle time (*best_cycle_time*)
best_assignment ← initial assignment;
best_cycle_time ← calculate_cycle_time(*best_assignment*);
neighborhood_list ← [1, 2, 3, 4] ; // Define 4 neighborhoods
$k \leftarrow 1$;
for *iteration* ← 1 **to** *max_iterations* **do**
  if $k > 4$ **then**
    $k \leftarrow 1$; // Reset to first neighborhood
  Apply neighborhood operation k to generate new assignment:;
  switch k **do**
    case *1* **do**
      Swap two tasks on the same machine
    case *2* **do**
      Move a task between non-adjacent machines
    case *3* **do**
      Move a task between adjacent machines
    case *4* **do**
      Shift a task from overloaded to underloaded machine
  if *new assignment satisfies all constraints* **then**
    new_cycle_time ← calculate_cycle_time(new assignment);
    if *new_cycle_time* < *best_cycle_time* **then**
      best_assignment ← new assignment;
      best_cycle_time ← new_cycle_time;
      $k \leftarrow 1$; // Restart from first neighborhood
    else
      $k \leftarrow k + 1$; // Move to next neighborhood
  else
    $k \leftarrow k + 1$; // Skip infeasible and change neighborhood
return *best_assignment, best_cycle_time*;

5. Computational experiments and case study

This section details the computational experiments and a real-world case study involving the production of engine cylinder heads, using data provided by our industrial partner. We present the key parameters of the dataset and discuss the results obtained from both the developed mathematical model and the proposed heuristic approach across operational, tactical, and strategic configuration levels. Given the absence of established benchmark instances for evaluating the performance of transfer line balancing models and heuristics across varying problem sizes (small, medium, and large), we have also developed a novel mathematical formulation to generate a standardized dataset for this specific problem domain. The mathematical models were solved using CPLEX 12.9, and the heuristic was coded in Python. All tests were executed on a Core i9 laptop with a 2.5 GHz CPU and 64 GB of RAM.

5.1. Case study and results discussion

This section presents a discussion of the computational performance of the developed mathematical model and the proposed heuristic algorithm in the context of a real-world case study. While the case is based on the operations of an industrial partner producing engine cylinder heads on a transfer line, the data used are surrogate representations that preserve the structural and operational complexity of the original system without disclosing proprietary information. The case is characterized by the following key parameters: NT = 118, NM = 9, NTO = 77, NF = 9, NS = 6, NOR = 35, TOMAG = 112, Precedence = 108, Zonp = 11, Zonn = 0, and Ter = 3. The average task–machine accessibility rates for the operational, tactical, and strategic configuration levels are $Acc1 \approx 13\%$, $Acc2 \approx 58\%$, and $Acc3 = 100\%$, respectively.

Table 9
Machine performance indicators at the operational configuration level.

Machine	Util. (%)	Idle (s)	Tasks	Busy (s)
Machine 1	92.2	69.0	14	813.0
Machine 2	76.8	204.8	27	677.2
Machine 3	78.7	187.6	14	694.4
Machine 4	79.8	178.4	10	703.6
Machine 5	71.8	249.0	10	633.0
Machine 6	80.3	173.5	13	708.5
Machine 7	60.6	347.3	16	534.7
Machine 8	70.4	261.4	5	620.6
Machine 9	87.3	112.4	9	769.6
Avg./Total	**77.5**	**1783.4**	**118**	–
Imbalance Index: 31.6%	Cycle Time: 882 s			

The mathematical model for the operational level comprises 5,431,481 variables and 15,098,034 constraints. Subject to a time limit of 7200 s, it yielded a cycle time of 889 s with an 8% optimality gap. The tactical model comprises 5,432,251 variables and 15,099,224 constraints but failed to find any feasible solution within 10,800 s. Similarly, the strategic model, comprising 5,432,351 variables and 15,106,124 constraints, found no solution within the time limit. These results highlight the computational limitations of the mathematical model at higher configuration levels.

In contrast, the proposed heuristic efficiently generated feasible solutions across all three levels. At the operational level, it found a solution in 2.24 s with a cycle time of 882 s. The Gantt chart in Fig. 1 depicts task assignments at this level, and the corresponding machine-level performance indicators are reported in Table 9. The following performance indicators are reported for each machine (j). *Busy time* is the total processing time of all tasks assigned to the machine. *Utilization* is the ratio of busy time to the cycle time, expressed as a percentage:

$$\text{Utilization}_j(\%) = \frac{\text{Busy Time}_j}{\text{Cycle Time}} \times 100 \tag{28}$$

Idle time is the difference between the cycle time and the busy time of each machine:

$$\text{Idle Time}_j = \text{Cycle Time} - \text{Busy Time}_j \tag{29}$$

The *workload imbalance index* measures the overall spread of busy times across machines relative to the cycle time:

$$\text{Imbalance Index}(\%) = \frac{\max_j(\text{Busy Time}_j) - \min_j(\text{Busy Time}_j)}{\text{Cycle Time}} \times 100 \tag{30}$$

A higher imbalance index indicates a more uneven workload distribution, with values approaching zero indicating a perfectly balanced line.

At the operational level, average machine utilization is 77.5%, with Machine 1 being the most loaded (92.2%) and Machine 7 the least loaded (60.6%). The imbalance index of 31.6% and total idle time of 1783.4 s indicate a notable workload disparity, with Machine 7 carrying only 16 tasks at 534.7 s of busy time, making it the primary candidate for workload rebalancing at higher configuration levels.

At the tactical level, the local search found a feasible solution in 124.62 s, reducing the cycle time to 798 s, a 9.5% improvement over the operational level. Fig. 2 shows the corresponding Gantt chart, and Table 10 reports the machine-level performance at this level.

Compared to the operational level, the tactical configuration yields a substantial improvement in workload balance. Average utilization rises from 77.5% to 85.6%, total idle time falls from 1783.4 s to 1035.8 s, and the imbalance index drops from 31.6% to 16.0%. Machine 7, previously the most underloaded machine, increases its utilization from 60.6% to 76.3%, demonstrating that tool reallocation effectively redistributes workload across machines.

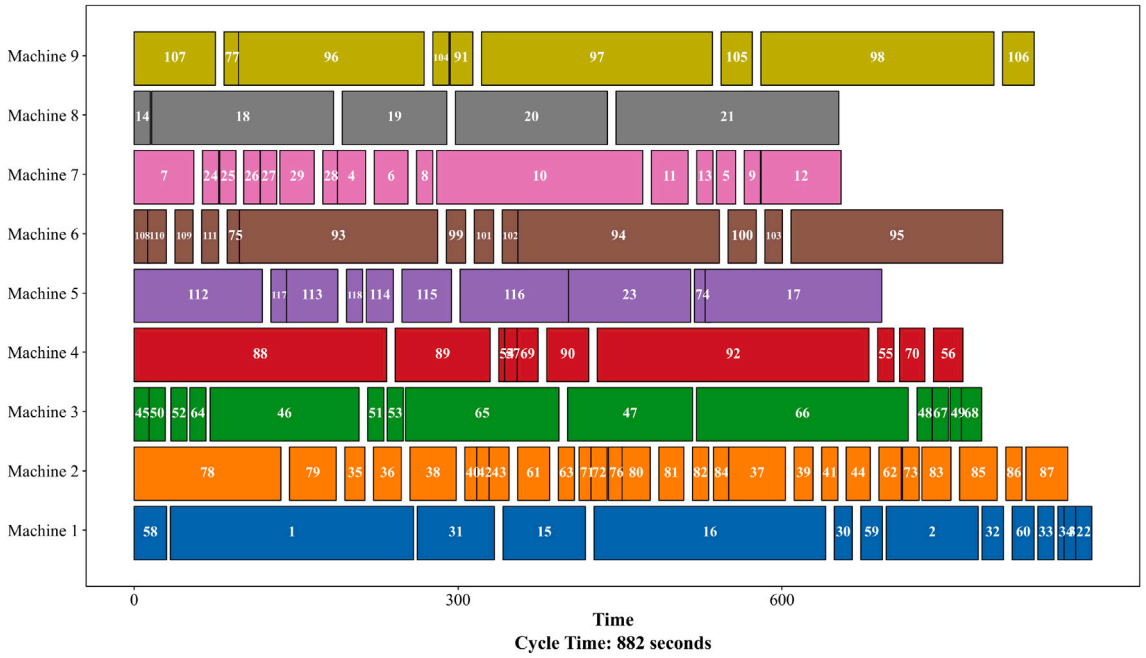


Fig. 1. Gantt chart of task assignments at the operational configuration level (Cycle Time: 882 s).

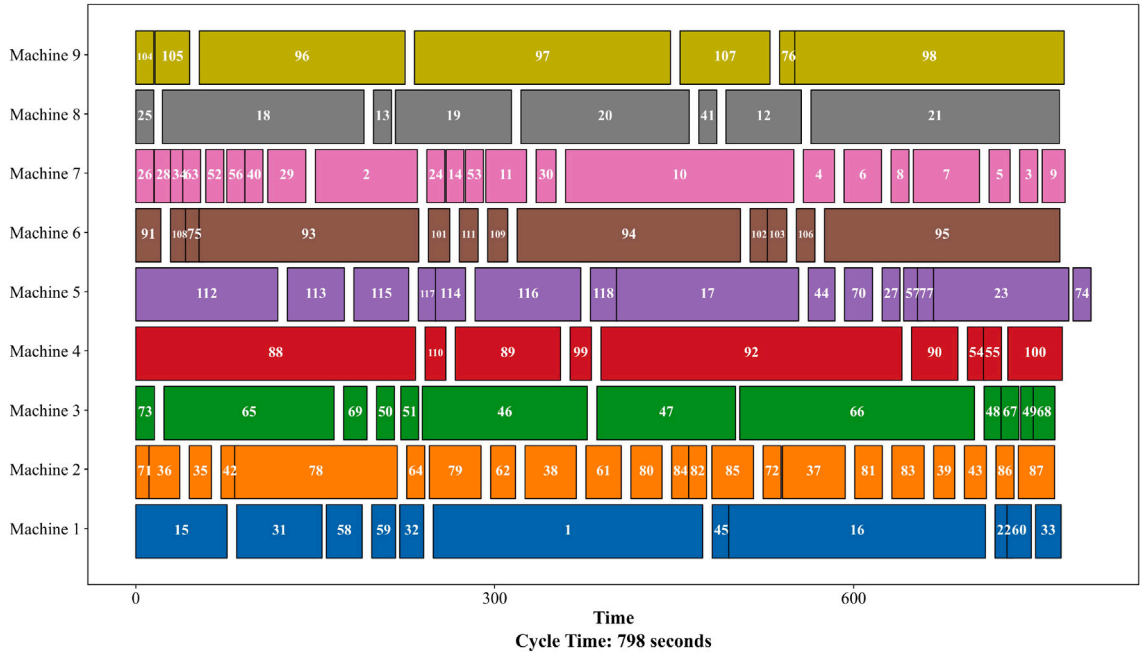


Fig. 2. Gantt chart of task assignments at the tactical configuration level (Cycle Time: 798 s).

At the strategic level, the local search yielded a feasible solution in 241.24 s, achieving a cycle time of 780 s, a further 2.3% improvement over the tactical level and an 11.6% improvement over the operational level. Fig. 3 and Table 11 present the results at this level.

The strategic configuration further improves average utilization to 87.4% and reduces total idle time to 883.5 s. The imbalance index declines only marginally from 16.0% to 15.8%, suggesting that most of the balance improvement was already achieved at the tactical level. This pattern of diminishing returns confirms that tactical reconfiguration through tool reallocation captures the majority of the available efficiency gain, while strategic reconfiguration involving simultaneous tool and fixture reassignment offers only incremental further improvement at considerably higher implementation cost and disruption.

5.2. Benchmark generation

To facilitate a rigorous evaluation of solution methods for the TLBP, we developed a mathematically grounded benchmark generator that systematically creates instances reflecting real-world operational, tactical, and strategic constraints. Unlike traditional random instance generation, our approach employs a two-stage procedure: a random pre-processing stage that generates equipment compatibility matrices, followed by a MILP-based optimization stage that determines

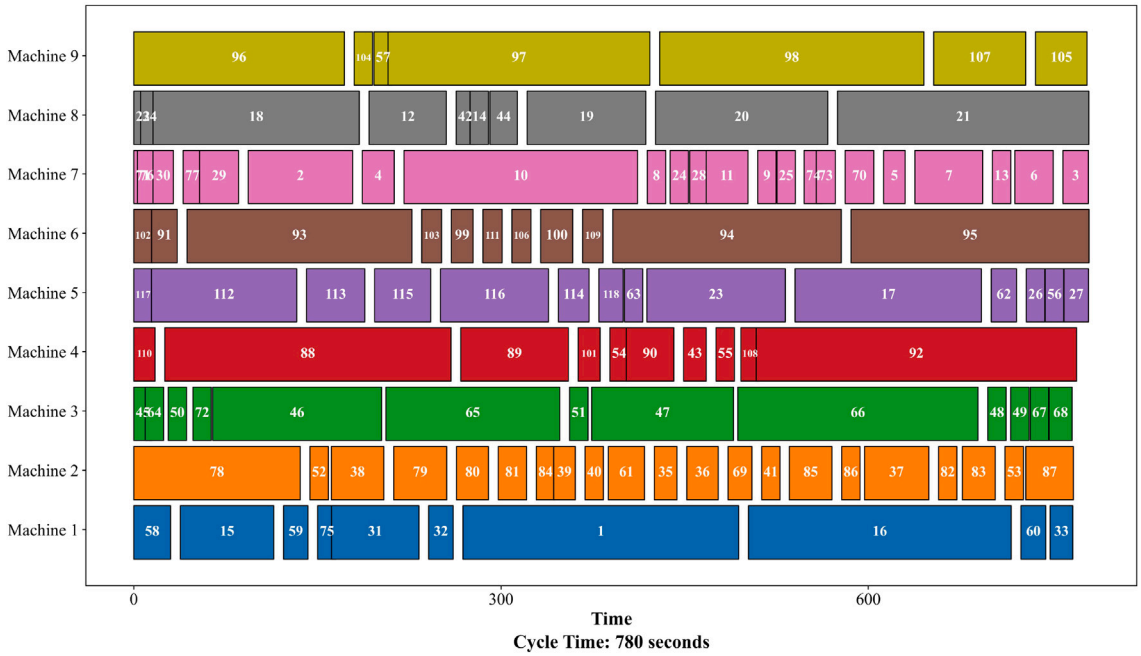


Fig. 3. Gantt chart of task assignments at the strategic configuration level (Cycle Time: 780 s).

Table 10
Machine performance indicators at the tactical configuration level.

Machine	Util. (%)	Idle (s)	Tasks	Busy (s)
Machine 1	88.6	91.1	11	706.9
Machine 2	75.2	198.0	22	600.0
Machine 3	86.9	104.3	12	693.7
Machine 4	89.0	87.6	9	710.4
Machine 5	87.5	100.0	15	698.0
Machine 6	85.8	113.6	12	684.4
Machine 7	76.3	189.2	22	608.8
Machine 8	89.7	82.0	8	716.0
Machine 9	91.2	70.0	7	728.0
Avg./Total	**85.6**	**1035.8**	**118**	–
Imbalance Index: 16.0%	Cycle Time: 798 s			

Table 11
Machine performance indicators at the strategic configuration level.

Machine	Util. (%)	Idle (s)	Tasks	Busy (s)
Machine 1	89.7	80.5	10	699.5
Machine 2	77.9	172.5	21	607.5
Machine 3	87.7	96.0	13	684.0
Machine 4	89.5	81.9	10	698.1
Machine 5	87.7	96.1	14	683.9
Machine 6	89.8	79.7	11	700.3
Machine 7	79.5	160.2	22	619.8
Machine 8	91.4	67.3	10	712.7
Machine 9	93.7	49.3	7	730.7
Avg./Total	**87.4**	**883.5**	**118**	–
Imbalance Index: 15.8%	Cycle Time: 780 s			

task–machine accessibility under controlled structural constraints. The complete mathematical formulation of the generator, including all notation, sets, parameters, decision variables, and constraints, is provided in Appendix B. The generator produces scalable instances in small, medium, and large sizes, each with controlled structural variations, as summarized in Table 12.

Stage 1: Equipment compatibility pre-processing. Three binary matrices capturing the physical equipment configuration of the transfer line are generated before the MILP is solved. The tool-task matrix ($ToTa_{ti}$) assigns exactly one tool to each task and then redistributes overloaded tool assignments to ensure that no tool serves more than $TOFR$ tasks, reflecting realistic tool magazine loading limits. The fixture-side matrix ($FiSi_{fs}$) assigns exactly $MaxNS$ sides to each fixture while guaranteeing that every side is covered by at least one fixture, consistent with rotary table configurations in real CNC lines. The machine-fixture matrix (MF_{jf}) enforces a strict one-to-one assignment between machines and fixtures, reflecting the fixture mounting constraint observed in practice.

Stage 2: MILP-based accessibility generation. Taking the three matrices from Stage 1 as fixed parameters, a MILP model jointly optimizes the side-task assignment ($SiTa_{si}$), the fixture-task assignment ($FiTa_{fi}$), and the machine–tool assignment (MTo_{ji}) to *maximize* total task–machine accessibility. Three accessibility matrices are then derived from the MILP solution, each corresponding to a different reconfiguration horizon. Under the operational horizon, $Acc1$ (≈13%) restricts accessibility to machines on which both the required tool and the required fixture are currently assigned (logical AND of $MT1_{ji}$ and $MT2_{ji}$), yielding the most constrained configuration. Under the tactical horizon, $Acc2$ (≈58%) considers a task accessible on any machine to which the required fixture is assigned ($MT2_{ji} = 1$), assuming tools can be reallocated across machines without changing fixture assignments. Under the strategic horizon, $Acc3$ (100%) treats all task–machine combinations as accessible, as both tools and fixtures may be freely reassigned in a full reconfiguration scenario. Three explicit constraints in the MILP generator jointly ensure that the accessibility structure of every produced instance permits a valid assignment of all tasks to machines. Constraint (B.20) requires that for every precedence pair (i, i'), the earliest machine accessible to task i does not exceed that of task i', so that no precedence chain becomes infeasible due to tool or fixture incompatibilities. Constraint (B.21) requires that every task is accessible on at least one machine. Constraint (B.22) requires that every machine has at least $MinNT$ accessible tasks.

Structural diversity is achieved through a deliberate two-dimensional variation scheme. For each of the 15 base problems, five machine-count variations are created (NM ∈ {3, 4, 5, 6, 7} for most instances). Within each variation, three constraint configurations are applied by varying zoning density (Zonp, Zonn) and technological constraint density (Ter) at three levels (approximately 17%, 33%, and 58% for zoning; 7%, 13%, and 24% for Ter), as detailed in Table 12. All post-MILP constraints, namely zoning inclusions ($Zp_{i,i'}$), zoning exclusions

Table 12
Key parameters for the benchmark generation.

Size	Problem	NT	NM	NTO	NS	NF	NOR	Zonp	Zonn	Ter
Small	Jackson	11	3, 4, 5, 6, 7	8	6	3, 4, 5, 6, 7	9, 8, 7, 6, 5	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Mitchell	21	4, 5, 6, 7, 8	14	6	4, 5, 6, 7, 8	18, 17, 16, 15, 14	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Sawyer	30	7, 8, 9, 10, 11	20	6	7, 8, 9, 10, 11	24, 23, 22, 21, 20	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Gunther	35	6, 7, 8, 9, 10	23	6	6, 7, 8, 9, 10	69, 77, 77, 77, 77	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Kilbrid	45	3, 4, 5, 6, 7	30	6	3, 4, 5, 6, 7	82, 87, 87, 87, 87	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
Medium	Hann	53	3, 4, 5, 6, 7	35	6	3, 4, 5, 6, 7	64, 64, 64, 64, 69	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Warnecke	58	3, 4, 5, 6, 7	38	6	3, 4, 5, 6, 7	69, 77, 77, 77, 77	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	TONGE70	70	3, 4, 5, 6, 7	46	6	3, 4, 5, 6, 7	68, 67, 66, 65, 64	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Wee-mag	75	3, 4, 5, 6, 7	49	6	3, 4, 5, 6, 7	73, 72, 71, 70, 69	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Arcus1	83	3, 4, 5, 6, 7	55	6	3, 4, 5, 6, 7	81, 80, 79, 78, 77	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
Large	Lutz3	89	3, 4, 5, 6, 7	59	6	3, 4, 5, 6, 7	87, 86, 85, 84, 83	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Mukherje	94	3, 4, 5, 6, 7	62	6	3, 4, 5, 6, 7	92, 91, 90, 89, 88	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	Arcus2	111	3, 4, 5, 6, 7	73	6	3, 4, 5, 6, 7	109, 108, 107, 106, 105	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	BARTHOLD	148	3, 4, 5, 6, 7	97	6	3, 4, 5, 6, 7	146, 145, 144, 143, 142	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%
	SCHOLL	297	3, 4, 5, 6, 7	194	6	3, 4, 5, 6, 7	295, 294, 293, 292, 291	17%, 33%, 58%	17%, 33%, 58%	7%, 13%, 24%

($Zn_{i,t'}$), technological requirements ($Ter_{i,j}$), and task–fixture incompatibility ($Pre_t_f_{i,f}$), are assigned using the $Acc1$ matrix as a filter, so that only task–machine combinations that are already feasible under the operational configuration are constrained. Secondary parameters such as spindle movement times, tool change times, table rotation times, and loading times are drawn from controlled discrete uniform distributions calibrated to realistic CNC machining ranges. Full details are provided in the publicly available dataset at https://figshare.com/s/3a46296855b7a2cb57c0. This scheme yields $5 \times 3 = 15$ instances per base problem and 225 instances in total.

The benchmarks reflect industrial conditions in two complementary ways. First, task precedence structures and processing times are taken directly from the ALBP benchmark library of Scholl et al. (1995), which is derived from real production environments. This ensures that the task dependency structures used in the experiments are industrially authentic across all problem sizes. Second, the TLBP-specific parameters are calibrated to match the real CNC transfer line described in Section 3. These include tool magazine slot capacities, fixture-side compatibility, one-to-one machine-fixture assignments, spindle movement times, and table rotation times, the latter computed as the absolute difference between side indices. Finally, the five levels of NM per base problem mirror the practical range of transfer line sizes encountered across operational, tactical, and strategic planning horizons.

5.3. Model versus heuristic comparison across configuration levels

This section presents a comparative analysis of the performance of the mathematical model and heuristic algorithm on 225 test instances generated using the proposed benchmark dataset. For each configuration level, the 225 test instances comprise 75 small-, 75 medium-, and 75 large-sized instances generated under different combinations of the problem parameters Zonp, Zonn, and Ter. The relative percentage deviation (Dev(%)) was calculated to evaluate the solution quality of each approach in comparison to the best-known solutions (BKS). The deviation is defined as:

$$\text{Dev}(\%) = \left(\frac{\text{SOL} - \text{BKS}}{\text{BKS}}\right) \times 100, \tag{31}$$

where SOL represents the solution value obtained by a given algorithm.

The mathematical model was solved using CPLEX with a computational time limit of 3600 s per instance. The heuristic algorithm was evaluated based on its computational efficiency and ability to achieve near-optimal solutions across all configuration levels (operational, tactical, and strategic), as discussed below.

5.3.1. Operational configuration level

Table 13 compares the performance of the mathematical model and the heuristic at the operational configuration level for three parameter settings (Zonp, Zonn, Ter). For each setting, the table reports how many instances the model solved to proven optimality, how many were solved without optimality proof, and how many failed, together with average optimality gaps, computation times, and deviations from the best-known solutions (BKS).

Across all settings, the mathematical model solved only about one-third of the 75 instances to optimality and roughly half to non-optimal feasible solutions, with 10–11 failures per configuration. Average optimality gaps for unsolved instances ranged between 3.5% and 4.56%, and average computation times were around 2370–2440 s per instance, although the deviation from the BKS remained very small (0.09%–0.13%). In contrast, the heuristic solved all 75 instances in every configuration, with average runtimes between 4.14 and 5.89 s and average deviations from the BKS between 0.25% and 0.31%. These results confirm that the heuristic provides high-quality solutions with substantially better scalability and computational efficiency than the exact model at the operational level. Additional details and experimental results can be accessed at: https://figshare.com/s/211389a364b70323a40a.

5.3.2. Tactical and strategic configuration levels

At the tactical and strategic configuration levels, the limitations of the exact model become more pronounced. The model was able to solve only the 10 smallest test instances (two small-sized problems under the different zoning configurations) and failed to find any feasible solution within the time limit for the remaining 65 instances. Consequently, the

Table 13
Model versus heuristic at the operational level.

Configuration parameters			Model						Heuristic		
			Solved			Average			Average		
Zonp	Zonn	Ter	Optimal	Not optimal	Fail	Gap (%)	Time (s)	Dev (%)	Solved	Time (s)	Dev (%)
17%	17%	7%	26	38	11	3.62	2392.47	0.09	75	5.89	0.25
33%	33%	13%	27	38	10	3.5	2368.64	0.13	75	4.14	0.29
58%	58%	24%	26	39	10	4.56	2438.67	0.11	75	4.27	0.31

Table 14
Model versus heuristic at the tactical level.

Configuration Parameters			Model						Heuristic		
			Solved			Average			Average		
Zonp	Zonn	Ter	Optimal	Not optimal	Fail	Gap (%)	Time (s)	Dev (%)	Solved	Time (s)	Dev (%)
17%	17%	7%	10	0	65	0.00	6.29	0.00	75	66.37	0.12
33%	33%	13%	10	0	65	0.00	4.70	0.00	75	63.75	0.04
58%	58%	24%	10	0	65	0.00	0.91	0.00	75	61.61	0.38

Table 15
Model versus heuristic at the strategic level.

Configuration parameters			Model						Heuristic		
			Solved			Average			Average		
Zonp	Zonn	Ter	Optimal	Not optimal	Fail	Gap (%)	Time (s)	Dev (%)	Solved	Time (s)	Dev (%)
17%	17%	7%	10	0	65	0.00	203.64	0.00	75	108.40	0.27
33%	33%	13%	10	0	65	0.00	35.95	0.00	75	106.78	0.04
58%	58%	24%	10	0	65	0.00	2.00	0.00	75	105.63	0.00

model-based performance indicators in Tables 14 and 15 are reported only for these 10 smallest instances, and the solution times are highly variable across zoning settings.

The heuristic, on the other hand, successfully solved all 75 instances at both tactical and strategic levels for all zoning configurations. At the tactical level, average runtimes ranged from approximately 61.61 to 66.37 s with average deviations from the BKS between 0.04% and 0.38%. At the strategic level, average runtimes ranged from 105.63 to 108.40 s with average deviations from the BKS between 0.00% and 0.27%. These results highlight that, beyond very small instances, the exact model is not scalable, whereas the proposed heuristic remains computationally efficient and robust across all problem sizes and configuration levels. Detailed results of the heuristic for all 75 instances at both the tactical and strategic levels are provided in https://figshare.com/s/2f069f538b4c7702149b.

5.3.3. Impact of configuration levels on cycle time

Table 16 presents the average cycle time results obtained under the operational, tactical, and strategic configuration levels for small-, medium-, and large-sized problem sets. It also reports the relative operational cycle time gaps with respect to the tactical and strategic configurations, calculated as follows:

$$\mathrm{RCG}_l(\%) = \left(\frac{\mathrm{CT_{Op}} - \mathrm{CT}_l}{\mathrm{CT}_l}\right) \times 100, \qquad l \in \{\mathrm{Ta}, \mathrm{St}\}, \tag{32}$$

where $\mathrm{CT_{Op}}$ denotes the cycle time at the operational level, and CT_l denotes the cycle time at configuration level l, with Ta and St representing the tactical and strategic levels, respectively. A positive value of RCG_l indicates the percentage by which the operational cycle time exceeds the cycle time achieved at the corresponding configuration level. Due to the computational complexity of solving all 225 test instances across the three configuration levels using the mathematical model, this analysis was conducted using the heuristic algorithm, which is capable of handling large-scale problems efficiently.

According to Table 16, for small-sized problems, the operational cycle time is, on average, 11.87% higher than the tactical cycle time and 12.65% higher than the strategic cycle time. For medium-sized problems, the corresponding relative gaps are 6.61% and 7.05%, while for large-sized problems they are 7.50% and 8.33%, respectively. These results indicate that expanding configuration decisions to include tool assignments at the tactical level and both tool and fixture assignments at the strategic level consistently leads to lower cycle times than the operational configuration. However, the differences between the tactical and strategic gap indicators are only 0.78, 0.44, and 0.83 percentage points for small-, medium-, and large-sized problems, respectively. This pattern indicates that most of the cycle time advantage is already achieved at the tactical level, while strategic reconfiguration provides only limited additional benefit relative to its higher implementation cost and disruption.

Table 16
Average cycle time and relative operational cycle time gaps across configuration levels.

Problem	Avg. cycle time			RCG (%)	
	Operational	Tactical	Strategic	Op vs. Ta	Op vs. St
Small					
Jackson	54.6	53.0	53.0	3.02	3.02
Mitchell	87.4	81.8	81.8	6.85	6.85
Sawyer	135.0	109.6	107.2	23.18	25.93
Gunther	174.0	147.0	145.6	18.37	19.51
Kilbrid	263.8	244.4	244.4	7.94	7.94
Average	142.96	127.16	126.40	**11.87**	**12.65**
Medium					
Hann	4868.0	4850.2	4850.2	0.37	0.37
Warnecke	557.2	501.2	501.0	11.17	11.22
Tonge	1231.2	1109.6	1088.4	10.96	13.12
Wee-mag	556.4	524.2	524.2	6.14	6.14
Arcus1	21951.8	21022.8	21022.8	4.42	4.42
Average	5832.92	5601.60	5597.32	**6.61**	**7.05**
Large					
Lutz3	626.6	609.2	609.2	2.86	2.86
Mukherje	1427.2	1289.4	1276.2	10.69	11.83
Arcus2	51484.6	46112.0	45423.4	11.65	13.34
Barthold	1863.0	1727.4	1711.2	7.85	8.87
Scholl	19491.4	18662.2	18607.0	4.44	4.75
Average	14978.56	13680.04	13525.40	**7.50**	**8.33**

Table 17
Sensitivity analysis on accessibility rate (Model vs. Heuristic).

Test case	Model						Heuristic			
	Acc 30%			Acc 60%			Acc 30%		Acc 60%	
	Gap (%)	Time (s)	Dev (%)	Gap (%)	Time (s)	Dev (%)	Time (s)	Dev (%)	Time (s)	Dev (%)
Kilbrid	7.61	3600	0.00	25.436	3600	1.71	0.1075	0.55	0.099	0.00
Hann	0	1721.6	0.03	9.467	3600	7.99	0.114	0.00	0.1187	0.00
Tonge70	7.4052	3600	0.00	11.251	3600	4	0.297	0.19	0.3016	0.00
Wee-mag	21.566	3600	0.00	40.63	3600	25.273	0.413	0.77	0.427	0.00

5.4. Sensitivity analyses

Two sensitivity analyses were conducted to evaluate the impact of key problem parameters on solution performance and computational behavior. Specifically, we examined: (1) the impact of task–machine accessibility on the performance of both the mathematical model and the heuristic algorithm, and (2) the influence of task processing time variability on solution quality and computational efficiency.

5.4.1. Impact of task–machine accessibility on solution performance

This analysis examines how varying levels of task–machine accessibility (Acc) impact the computational performance and solution quality of both the mathematical model and the proposed heuristic. All other settings were kept fixed, and only the accessibility rate was varied (30% and 60%) for four representative test problems: Kilbrid, Hann, Tonge70, and Wee-mag. The results are summarized in Table 17.

The mathematical model demonstrated a clear sensitivity to the accessibility rate. In the 30% accessibility scenario, the model achieved moderate optimality gaps (7.61% for Kilbrid, 7.41% for Tonge70, and 21.57% for Wee-mag), and solved the Hann instance optimally (0% gap) in 1721.6 s. However, when accessibility increased to 60%, the model's performance declined significantly, with higher optimality gaps (25.44% for Kilbrid, 11.25% for Tonge70, 40.63% for Wee-mag, and 9.47% for Hann), while still hitting the 3600-second time limit in most cases. For all instances except Hann at 30%, the model reached the time limit of 3600 s. This result suggests that higher accessibility introduces a larger solution space, which increases computational complexity for the model, making it harder to converge to optimal or near-optimal solutions within the time limit.

In contrast, the heuristic remained robust and efficient across both accessibility levels. All test instances were solved in under 0.5 s, with negligible changes in performance between the two scenarios. For example, Kilbrid was solved in 0.108 s at 30% accessibility and 0.099 s at 60%, while Tonge70 remained around 0.3 s in both settings. The deviation from the best-known solution (Dev %) remained low for the heuristic in all cases, never exceeding 0.77% at 30% accessibility and reaching 0% for all instances at 60% accessibility. These findings highlight the heuristic's ability to handle expanded solution spaces without a significant computational penalty.

Overall, this analysis confirms that while increased accessibility can complicate the exact model's search process, it does not adversely affect the heuristic. This reinforces the suitability of the heuristic approach for large-scale or highly flexible production environments where task–machine compatibility is broader.

5.4.2. Impact of task time variability on computational behavior

A sensitivity analysis was conducted to investigate the impact of changes in task time variability on two key outcomes: the optimality gap and the time required to solve the problem. This analysis kept all settings fixed while adjusting the task time variability range to [1, 100], [1, 500], and [1, 1000] for four representative test cases: Kilbrid, Hann, Tonge70, and Wee-mag. As shown in Table 18, increasing the variability range of task times generally resulted in a clear decrease in the optimality gap across all cases. For example, in the Kilbrid case, the gap decreased from 3.711% with a [1,100] range to only 0.088% with [1,1000]. Similar patterns were observed in Tonge70 and Wee-mag, where the gaps decreased from 5.528% and 9.86% to 0.387% and 0.944%, respectively. These results suggest that having a wider spread in task times makes the problem easier to solve and leads to better-quality solutions. Interestingly, although most cases reached the 3600-second time limit, the Hann case with ranges [1,500] and [1,1000] was solved much faster (38.64 s and 24.108 s), showing that some problem types may benefit even more from greater time variation. The deviation (Dev %) for the model remained consistently very low or zero, indicating stable and reliable performance across all variability ranges, with only minor fluctuation in a few cases, such as Wee-mag at the highest range. Similarly, the heuristic's deviation stayed minimal, always below 1%, highlighting its consistent and robust performance regardless of task time variability. Overall, this analysis shows that broader task time ranges can help improve both solution quality and solving efficiency. The heuristic algorithm, in contrast, solved all instances consistently in under 0.5 s, regardless of the task time range. Its performance remained stable and efficient across all cases, further demonstrating the heuristic's robustness and suitability for handling diverse task time distributions with minimal computational effort.

6. Managerial insights

The results show that configuration flexibility should be adopted selectively rather than uniformly across all transfer lines. Table 19 consolidates the findings into a decision-support framework that links each configuration level to its corresponding reconfiguration action, implementation effort, investment requirement, and observed relative operational cycle time gap.

The largest relative operational cycle time gaps are observed for small-sized instances. In these cases, the operational cycle time is, on average, 11.87% higher than the tactical cycle time and 12.65% higher than the strategic cycle time. For medium-sized instances, the corresponding relative gaps are 6.61% and 7.05%, while for large-sized instances they are 7.50% and 8.33%, respectively. These results show that the magnitude of the cycle time advantage varies with problem structure and is largest, on average, for the small-sized instances considered in this study.

Three implications follow from this framework. First, the tactical level is sufficient in many cases. Allowing tool reallocation across machines captures most of the benefit associated with broader configuration flexibility while avoiding the greater cost, disruption, and implementation effort of fixture-level reconfiguration. The differences between the tactical and strategic gap indicators are only 0.78 percentage points (p.p.) for small-sized instances, 0.44 p.p. for medium-sized instances, and 0.83 p.p. for large-sized instances. This suggests that, unless fixture changes are already planned or a broader long-term reconfiguration is justified, the tactical level is likely to be the more economical managerial choice.

Second, the results indicate a clear pattern of diminishing returns between the tactical and strategic levels. Most of the cycle time advantage over the operational configuration is achieved once accessibility is expanded to the tactical level. Strategic configuration decisions further reduce cycle time, but the additional advantage is comparatively small. Managers should therefore reserve strategic reconfiguration for settings in which long-term product-mix changes, persistent bottlenecks, or

Table 18
Sensitivity analysis on task time variability (Model vs. Heuristic).

Test case	Task time range	Model			Heuristic	
		Gap (%)	Time (s)	Dev (%)	Time (s)	Dev (%)
Kilbrid	[1, 100]	3.711	3600	0.00	0.0783	0.65
	[1, 500]	1.564	3600	0.02	0.0624	0.00
	[1, 1000]	0.088	3600	0.00	0.0702	0.18
Hann	[1, 100]	0.018	3600	0.00	0.119	0.25
	[1, 500]	0.000	38.64	0.00	0.0963	0.00
	[1, 1000]	0.000	24.108	0.00	0.1308	0.00
Tonge70	[1, 100]	5.528	3600	0.00	0.2318	0.00
	[1, 500]	1.078	3600	0.00	0.364	0.00
	[1, 1000]	0.387	3600	0.00	0.268	0.12
Wee-mag	[1, 100]	9.860	3600	0.00	0.3787	0.00
	[1, 500]	2.004	3600	0.00	0.4417	0.013
	[1, 1000]	0.944	3600	0.039	0.355	0.00

Table 19
Decision-support framework for configuration level selection.

Criterion	Operational	Tactical	Strategic
Reconfiguration lever	Task reassignment under fixed tool and fixture configurations	Tool reallocation	Tool and fixture reallocation
Observed accessibility density	13%	58%	100%
RCG-Small	–	11.87%	12.65% (+0.78 p.p. vs. tactical)
RCG-Medium	–	6.61%	7.05% (+0.44 p.p. vs. tactical)
RCG-Large	–	7.50%	8.33% (+0.83 p.p. vs. tactical)
Implementation effort	Low	Moderate	High
Investment requirement	Minimal	Tool relocation or duplication and magazine-capacity verification	Fixture relocation or redesign and potential line downtime
Planning horizon	Days	Weeks to months	Months to a year
Recommended when	Tool relocation is infeasible because magazine capacity is full or tools are machine-specific	Spare magazine capacity exists and tools are transferable between machines	Fixture investment is already planned for product-mix or capacity reasons

major capacity pressures justify broader system changes, treating the additional cycle time benefit as a secondary rather than the primary justification.

Third, these findings suggest that accessibility density is an important consideration in configuration decisions. When accessibility is limited, increasing configuration flexibility can provide greater benefits by broadening the range of feasible task–machine assignments. However, once a moderate level of accessibility has been achieved, further increases in flexibility may yield smaller additional benefits. Managers should therefore prioritize measures that expand feasible accessibility, particularly through tool-related flexibility, before investing in more disruptive fixture-level reconfiguration.

7. Conclusion and future research

This study presented a comprehensive investigation into the TLBP, incorporating hierarchical configuration decisions across operational, tactical, and strategic levels. A novel MILP formulation was developed to capture a wide range of industrially relevant constraints, including precedence, technological compatibility, zoning, accessibility, tool and fixture limitations, and sequencing requirements. To overcome the computational limitations of exact optimization for large-scale problems, a constructive heuristic was proposed.

The findings demonstrate that expanding the scope of configuration decisions consistently leads to lower cycle times across the considered configuration levels. Notably, tactical-level tool reallocation and strategic-level tool–fixture reassignment provide substantial improvements in system performance without necessitating structural redesigns of the line. The proposed heuristic consistently produced high-quality solutions with short runtimes and substantially better scalability than the exact method, particularly for medium- and large-sized instances. This research contributes to both theory and practice by: (1) modeling and solving a practical, highly constrained TLBP with hierarchical configuration decisions in a real-world case; (2) proposing a scalable and effective solution method suitable for industrial implementation; and (3) generating various benchmark datasets and analyzing the impact of hierarchical configuration decisions, task–machine accessibility and task time variability on the resulting solutions.

Several limitations should be considered when interpreting these results. The model assumes a fixed serial-line topology in which machines are arranged sequentially without parallel stations, buffers, or alternative routing. This assumption reflects the configuration of the case company and maintains formulation tractability. However, it limits the available reconfiguration levers to task, tool, and fixture reassignment. In lines with parallel machines or flexible routing, additional improvements may be achieved through workload redistribution across alternative paths, and the relative value of tactical and strategic reconfiguration may differ.

The model is also deterministic and considers a single-product environment. Processing times, tool changes, and rotation times are assumed to be known constants, and all parts follow an identical task

set. Therefore, the reported cycle time advantages represent performance under stable operating conditions rather than expected performance under uncertainty or product-mix variation. The observed diminishing returns between tactical and strategic reconfiguration should consequently be interpreted within this context.

These limitations suggest several directions for future research. Introducing uncertainty in processing times, tool life, and machine availability would require stochastic or robust optimization approaches that minimize expected cycle time or control service-level and risk-based performance measures. Extending the model to mixed-model environments would require product-specific accessibility matrices and an expanded tool magazine constraint that considers the combined tool requirements of multiple products. Such settings may increase the importance of magazine capacity and potentially alter the trade-off between tactical and strategic reconfiguration. Extending the framework to parallel or reconfigurable line architectures would require reformulating the completion time constraints to capture workload sharing across alternative stations. Developing multi-objective formulations that balance cycle time against energy consumption, tooling cost, or maintenance requirements could further broaden the practical relevance of the framework.

CRediT authorship contribution statement

Pedram Beldar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Amir Nourmohammadi:** Writing – review & editing, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Masood Fathi:** Supervision, Project administration, Funding acquisition, Conceptualization, Investigation. **Amos H.C. Ng:** Supervision, Resources, Investigation.

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Appendix A. Linearization of first and last task identification

To account for loading and unloading times in Constraint (21), it is necessary to identify which task is first and which is last on each machine. This is achieved through the binary variable $fl_{ii'j}$, which equals 1 if task i is the first and task i' is the last task on machine j, and 0 otherwise. Its computation requires the following auxiliary variables:

$z1_{ii'j}$ = binary auxiliary variable

$z2_{ii'j}$ = binary auxiliary variable

$stdi_{ii'j} \in \mathbb{R}$, continuous auxiliary variable

The variable $stdi_{ii'j}$ captures the relative positional difference between tasks i and i' on machine j, encoded as a weighted combination of machine index and order position. Variables $z1_{ii'j}$ and $z2_{ii'j}$ use Big-M constraints to detect whether task i occupies the first position and task i' occupies the last position, respectively. The variable $fl_{ii'j}$ is then activated when both conditions hold simultaneously.

The Big-M parameter B is not set as an arbitrary constant but is derived from the structure of these linearization constraints. Since each task is assigned to exactly one machine-order pair, $stdi_{ii'j}$ takes values in the range $[-(10 \cdot NM + NOR),\ 10 \cdot NM + NOR]$. Combined with the maximum value of $(ntm_j - 1) \le NT - 1$, the minimum valid Big-M is:

$$B \ge 10 \cdot NM + NOR + NT$$

In this study, B is set according to this bound for each instance, ensuring the constraints are both valid and as tight as possible, thereby avoiding unnecessary relaxation of the feasible region. The small constant δ is set to 0.01 to enforce strict inequalities while maintaining numerical stability.

$$stdi_{ii'j} = \sum_{o \in O} (10 \cdot j + o) \cdot x_{ijo} - \sum_{o \in O} (10 \cdot j + o) \cdot x_{i'jo}, \quad \forall i, i' \in I, j \in J \tag{A.1}$$

$$B \cdot z1_{ii'j} \ge stdi_{ii'j} - \left(-(ntm_j - 1)\right) + \delta, \quad \forall i, i' \in I, j \in J \tag{A.2}$$

$$B \cdot (1 - z1_{ii'j}) \ge \left(-(ntm_j - 1)\right) - stdi_{ii'j} - \delta, \quad \forall i, i' \in I, j \in J \tag{A.3}$$

$$B \cdot z2_{ii'j} \ge \left(-(ntm_j - 1)\right) - stdi_{ii'j} + \delta, \quad \forall i, i' \in I, j \in J \tag{A.4}$$

$$B \cdot (1 - z2_{ii'j}) \ge stdi_{ii'j} - \left(-(ntm_j - 1)\right) - \delta, \quad \forall i, i' \in I, j \in J \tag{A.5}$$

$$fl_{ii'j} \ge z1_{ii'j} + z2_{ii'j} - 1, \quad \forall i, i' \in I, j \in J \tag{A.6}$$

Constraints (A.2)–(A.3) activate $z1_{ii'j} = 1$ when task i occupies the earliest position among all tasks on machine j. Constraints (A.4)–(A.5) activate $z2_{ii'j} = 1$ when task i' occupies the latest position. Constraint (A.6) sets $fl_{ii'j} = 1$ when both conditions hold, identifying the unique first–last task pair on each machine.

Appendix B. Benchmark generator: MILP formulation

This appendix presents the complete MILP formulation used in the MILP-based accessibility generation step of the benchmark generator described in Section 5.2. The model receives the pre-computed matrices $ToTa_{ti}$, $FiSi_{fs}$, and MF_{jf} from the pre-processing step as fixed parameters, and jointly determines the side-task assignment ($SiTa_{si}$) and machine–tool assignment (MTo_{jt}) that maximize total task–machine accessibility while guaranteeing feasibility across all configuration levels.

Sets:	
$I = \{1, \dots, NT\}$	Set of tasks indexed by i
$J = \{1, \dots, NM\}$	Set of machines indexed by j
$T = \{1, \dots, NTO\}$	Set of tools indexed by t
$F = \{1, \dots, NF\}$	Set of fixtures indexed by f
$S = \{1, \dots, NS\}$	Set of sides indexed by s
Parameters:	
NT	Number of tasks
NM	Number of machines
NTO	Number of tools
NS	Number of sides
NF	Number of fixtures
$MinNT$	Minimum number of tasks accessible per machine
$MaxNTO$	Minimum number of tools assigned per machine
$Pr_{i,i'}$	1 if task i must precede task i', 0 otherwise
$ToTa_{ti}$	1 if tool t is required by task i, 0 otherwise (pre-computed)
$FiSi_{fs}$	1 if fixture f covers side s, 0 otherwise (pre-computed)
MF_{jf}	1 if fixture f is mounted on machine j, 0 otherwise (pre-computed)
Decision variables:	
$MTo_{jt} \in \{0, 1\}$	1 if tool t is assigned to machine j
$SiTa_{si} \in \{0, 1\}$	1 if task i is assigned to side s

$FiTa_{fi} \in \{0,1\}$	1 if task i is compatible with fixture f
$MT1_{ji} \in \{0,1\}$	1 if machine j can access task i via its tools
$MT2_{ji} \in \{0,1\}$	1 if machine j can access task i via its fixture
$ACC_{ji} \in \{0,1\}$	1 if machine j can access task i via both tool and fixture
$min_ACC_i \in \mathbb{Z}^+$	Index of the lowest-numbered machine accessible to task i
$FiSiTa_{fsi} \in \{0,1\}$	Linearization auxiliary: fixture–side–task compatibility
$MToTa_{jti} \in \{0,1\}$	Linearization auxiliary: machine–tool–task compatibility
$MFiTa_{jfi} \in \{0,1\}$	Linearization auxiliary: machine–fixture–task compatibility

Objective:

$$\text{Maximize} \quad \sum_{j\in J}\sum_{i\in I} ACC_{ji} \tag{B.1}$$

Side-task assignment:

$$\sum_{s\in S} SiTa_{si} = 1, \quad \forall i \in I \tag{B.2}$$

Fixture-task compatibility (linearization of $FiSi_{fs} \cdot SiTa_{si}$):

$$FiSiTa_{fsi} \le FiSi_{fs}, \quad \forall f,s,i \tag{B.3}$$

$$FiSiTa_{fsi} \le SiTa_{si}, \quad \forall f,s,i \tag{B.4}$$

$$FiSiTa_{fsi} \ge FiSi_{fs} + SiTa_{si} - 1, \quad \forall f,s,i \tag{B.5}$$

$$FiTa_{fi} = \sum_{s\in S} FiSiTa_{fsi}, \quad \forall f,i \tag{B.6}$$

$$\sum_{f\in F} FiTa_{fi} = 1, \quad \forall i \in I \tag{B.7}$$

Machine-task accessibility by tool (linearization of $MTo_{jt} \cdot ToTa_{ti}$):

$$MToTa_{jti} \le MTo_{jt}, \quad \forall j,t,i \tag{B.8}$$

$$MToTa_{jti} \le ToTa_{ti}, \quad \forall j,t,i \tag{B.9}$$

$$MToTa_{jti} \ge MTo_{jt} + ToTa_{ti} - 1, \quad \forall j,t,i \tag{B.10}$$

$$MT1_{ji} = \sum_{t\in T} MToTa_{jti}, \quad \forall j,i \tag{B.11}$$

Machine-task accessibility by fixture (linearization of $MF_{jf} \cdot FiTa_{fi}$):

$$MFiTa_{jfi} \le MF_{jf}, \quad \forall j,f,i \tag{B.12}$$

$$MFiTa_{jfi} \le FiTa_{fi}, \quad \forall j,f,i \tag{B.13}$$

$$MFiTa_{jfi} \ge MF_{jf} + FiTa_{fi} - 1, \quad \forall j,f,i \tag{B.14}$$

$$MT2_{ji} = \sum_{f\in F} MFiTa_{jfi}, \quad \forall j,i \tag{B.15}$$

Combined accessibility:

$$ACC_{ji} \le MT1_{ji}, \quad \forall j,i \tag{B.16}$$

$$ACC_{ji} \le MT2_{ji}, \quad \forall j,i \tag{B.17}$$

Minimum accessible machine index:

$$min_ACC_i \le j + (1 - ACC_{ji}) \cdot |J|, \quad \forall i,j \tag{B.18}$$

$$min_ACC_i \ge j \cdot ACC_{ji}, \quad \forall i,j \tag{B.19}$$

Precedence-accessibility consistency:

$$min_ACC_i \le min_ACC_{i'}, \quad \forall i,i' \text{ if } Pr_{i,i'} = 1 \tag{B.20}$$

Feasibility requirements:

$$\sum_{j\in J} ACC_{ji} \ge 1, \quad \forall i \in I \tag{B.21}$$

$$\sum_{i\in I} ACC_{ji} \ge MinNT, \quad \forall j \in J \tag{B.22}$$

$$\sum_{t\in T} MTo_{jt} \ge MaxNTO, \quad \forall j \in J \tag{B.23}$$

The objective (B.1) maximizes total accessibility across all machine-task pairs, ensuring the generated instances are as rich as possible while satisfying all feasibility requirements. Constraint (B.2) assigns each task to exactly one machining side. Constraints (B.3)–(B.5) linearize the product $FiSi_{fs} \cdot SiTa_{si}$ to determine fixture-side-task compatibility. Constraint (B.6) aggregates this into the fixture-task compatibility variable $FiTa_{fi}$. Constraint (B.7) ensures each task is compatible with exactly one fixture. Constraints (B.8)–(B.10) linearize $MTo_{jt} \cdot ToTa_{ti}$ to determine machine–tool-task compatibility. Constraint (B.11) aggregates this into the machine-task accessibility by tool $MT1_{ji}$. Constraints (B.12)–(B.14) linearize $MF_{jf} \cdot FiTa_{fi}$ to determine machine-fixture-task compatibility. Constraint (B.15) aggregates this into the machine-task accessibility by fixture $MT2_{ji}$. Constraints (B.16)–(B.17) define ACC_{ji} as the logical AND of tool-based and fixture-based accessibility, producing the operational accessibility matrix $Acc1$. Constraints (B.18)–(B.19) compute min_ACC_i as the index of the lowest-numbered machine accessible to task i. Constraint (B.20) enforces that for every precedence pair (i,i'), the earliest accessible machine of task i does not exceed that of task i', preventing infeasible precedence chains. Constraints (B.21)–(B.22) guarantee that every task has at least one accessible machine and every machine has at least $MinNT$ accessible tasks. Constraint (B.23) ensures each machine is assigned at least $MaxNTO$ tools.

Data availability

Data will be made available on request.

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